

School of Arts and Sciences
Department of Computer Science and Information Technology
Summer 2024 – 2025

CSCI378 Data Structures and Algorithms

Class Day	: _____	Class Time:	: _____	Room:	: _____
Instructor	: _____				
E-mail	: _____				
Office	: _____	Office Hours:	: _____	Extension No:	: _____

A. Course Description

This course covers the design and implementation of important data structures and their algorithms in addition to their time complexity using the big-Oh notation. The data structures considered include stacks, queues, lists, linked lists, trees, and graphs. An approach based on abstract data types and classes will be emphasized and sorting algorithms are discussed.

B. Course Objectives

- CO-1 Explain different data structures and the algorithms that manipulate them
- CO-2 Test abstract data types through programming techniques
- CO-3 Identify the advantages and disadvantages of many data structures
- CO-4 Explain what algorithm analysis is, asymptotic notation and big-oh notation
- CO-5 Learn how to compute the time complexity of algorithms

C. Course Learning Outcomes

Outcome	Details
CLO-1:	Discover different linear (linked lists, stacks and queues) and non-linear (binary tree and graph) data structures with their basic operations
CLO-2:	Select appropriate data structure to be used in a specific program to solve real world problems
CLO-3:	Examine how the choice of data structures and their underpinning algorithms (e.g. iterative, recursion) affect the performance of programs
CLO-4:	Analyze the complexity of any algorithm to be used as a solution to a problem
CLO-5:	Compare between different sorting algorithms to determine the best one to use.

D. Outcomes Mapping Metrics

Course outcomes mapping to Program Outcomes:

Course Outcome	Program Outcomes								
	(PLO-1:)	(PLO-2:)	(PLO-3:)	(PLO-4:)	(PLO-5:)	(PLO-6:)	(PLO-7:)	(PLO-8:)	(PLO-9:)
CLO-1:	X								
CLO-2:			X						
CLO-3:	X		X	X					
CLO-4:	X	X	X						
CLO-5:			X						

E. Topics Covered

#	Topic Title
1	Linked List
2	Stacks and Queues
3	Algorithm Analysis
4	Recursion
5	Binary Trees
6	Graphs
7	Sorting Algorithms

F. Prerequisite - Corequisite

Prerequisite Courses	Corequisite Courses
CSCI300 - Intermediate Programming with Objects	

G. Course Material

Type	Title
Textbook	Data Structures in Java - Noel Kalicharan - 5 - 2008(ISBN: 143827517X)

H. Detailed Schedule

	Book reference	Topic Details
Week1 -session1 30/6	Data Structures in Java	A look of the syllabus + Overview of Data Structures Chapter 1: Linked List 1Introduction 1.1 Basic operations 1.1.1 Counting nodes 1.1.2 Searching 1.1.3 Printing the values
Week1 – session2 1/7	Data Structures in Java	Chapter 1: Linked List 1.2 Building a linked list (Tail) 1.3 Insertion 1.4 Building a linked list (Head) 1.5 Deletion 1.6 Sorted lists
Week1 – session3 2/7	Data Structures in Java	Chapter 1: Linked List 1.7 A linked list class 1.8 Palindrome 1.9 Merging lists Chapter 1 (page 33): Solve exercise 10 POST ASSIGNMENT1

Week1 -session4 3/7	Data Structures in Java	Chapter 2: Stacks and Queues 2.1 ADT 2.2 Stacks 2.2.1 Implementing a stack using an array 2.2.2 Implementing a stack using Linked Lists
Week2-session1 7/7	Data Structures in Java	Chapter 2: Stacks and Queues 2.3 A general stack type, 2.3.1 Example convert from decimal to binary 2.4.1 How to evaluate a postfix expression Chapter 2 (page 57): Solve exercise 10
Week2-session2 8/7	Data Structures in Java	Chapter 2: Stacks and Queues 2.5 Queues 2.5.1 Implementing a queue using an array 2.5.2 Implementing a queue using Linked Lists POST ASSIGNMENT 2
Week2-session3 9/7	Data Structures in Java	Algorithm Analysis Counting primitive operations.Measuring operations as a function of input size. Focusing on the worst-case input. Time complexity functions: Constant, Logarithm, Linear, N-Log-N, Quadratic, Cubic, Exponential. Comparing Growth Rates. Asymptotic Analysis: The big-Oh notation.
Week2-session4 10/7	Data Structures in Java	Solve exercises related to Algorithm Analysis POST ASSIGNMENT 3
Week3-session1 14/7	Midterm exam Thursday 14 July	Chapters 1 and 2
Week3-session2 15/7	Data Structures in Java	Recursion Non void recursive methods Solve Exercises Recursion Void recursive methods Recursion Solve Exercises
Week3-session3 16/7	Data Structures in Java	Recursion Void recursive methods Recursion Solve Exercises
Week3-session4 17/7	Data Structures in Java	Chapter 3: Binary Trees 3.1 Trees 3.2 Binary Trees 3.3 Traversing a binary tree 3.4 How to represent a binary tree
Week4-session1 21/7	Data Structures in Java	Chapter 3: Binary Trees 3.5 Binary Search Trees (BST), 3.9 Level-order traversal 3.10 Some useful tree functions 3.11 BST deletion
Week4-session2 22/7	Data Structures in Java	Chapter 3: Binary Trees Chapter 3 (pages 112-113): Solve exercises 5, 6, and 11 POST ASSIGNMENT 4
Week4-session3 23/7	Data Structures in Java	Chapter 5: Graphs 5.1 Graph Terminology 5.2 How to represent a graph 5.3 Building a graph [brief] 5.4 Traversing a graph 5.5 Depth-first traversal
Week4-session4 24/7	Test (60 minutes)	Algorithm Complexity +Recursion + Chapter 3 (till section 3.9)

Week5-session1 28/7	Data Structures in Java	Chapter 5: Graphs 5.7 Breadth-first traversal 5.8 Dijkstra's algorithm (without the implementation) Chapter 5 (page 196): Solve exercises 2 and 3 POST ASSIGNMENT5
Week5-session2 29/7	Data Structures in Java	<u>Chapter 4: Sorting Algorithms</u> Bubble-Sort, Merge-Sort, (Quick-sort), The running time of sorting algorithms
Week 5-session3 30/7	N/A	Review session
Week5-session4 31/7	N/A	Review session

I. Course Assessment

#	Activity	Percentage
1	Midterm Exam	35%
2	Test	30%
3	Final	35%

I. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A "10%" has been allocated on the "Marking Profile" for attendance. A student who hasn't missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam's grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

E-mails

Students should use their university e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your university e-mail accounts.

University Peer Review Policy

All classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4	D+	72	2.2	
65	1.5		73	2.3	B+
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9	B	87	3.7	
80	3		88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

	Program Outcomes
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category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6 for CSIT=> Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working